Statistics

Chapter 7 — Instructor Solutions Manual

October 12, 2026 • Total Exercises: 15

Chapter 7 **Exercise 1**

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PROBLEM STATEMENT

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Step 1

A probability is always between 0 and 1 (inclusive), or equivalently, between 0% and 100% (inclusive).

This then implies that a probability can **never** be greater than 1 or 100%.

Result

Never

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PROBLEM STATEMENT

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Step 1

An **experiment** deliberately imposes some treatment on individuals in order to observe their responses.

Moreover, an experiment observes and/or measures some activity.

Result

Experiment

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PROBLEM STATEMENT

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Step 1

The collection of one or multiple outcomes to an experiment is called an **event**.

Result

Event

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PROBLEM STATEMENT

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Step 1

The probability that two or more events will happen at the same time is called the **joint** probability.

Result

Joint

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PROBLEM STATEMENT

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Step 1

The order in which events are counted is important in a **permutation**, while the order is not important in a **combination**.

Result

(a) Permutation

(b) Combination

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PROBLEM STATEMENT

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Step 1

Each probability distribution requires that the total probability of all possible outcomes is equal to 1.

A discrete probability distribution then also requires that the sum of the (probability of) possible outcomes is equal to 1.

Result

1

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PROBLEM STATEMENT

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Step 1

The four conditions for a binomial setting are: binary (success/failure), independent trials, fixed number of trials and probability of success is the same for each trial.

We note that a binomial distribution requires that there are only two possible outcomes (a success or a failure) and thus "three or more outcomes" is not one of the requirements for a binomial distribution.

Result

Three or more outcomes

PROBLEM STATEMENT

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Step 1

The **normal probability distribution** is dependent on the value of the mean and the standard deviation.

However, this then implies that a different mean and/or a different standard deviation leads to a different normal distribution and thus there are many possible normal probability distributions.

Since there are infinitely many possible values for the mean and infinitely many possible values for the standard deviation, there are infinitely many possible normal distributions.

🎯 Result

Infinite

PROBLEM STATEMENT

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Step 1

The **normal probability distribution** is dependent on the value of the mean and the standard deviation.

However, this then implies that a different mean and/or a different standard deviation leads to a different normal distribution and thus there are many possible normal probability distributions.

Since there are infinitely many possible values for the mean and infinitely many possible values for the standard deviation, there are infinitely many possible normal distributions.

However, a standard normal distribution is the normal distribution with mean 0 and standard deviation 1, which implies that there is exactly 1 standard normal distribution.

🎯 Result

1

PROBLEM STATEMENT

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Step 1

Given:

$$P(-0.76 < Z < 0)$$

Determine the corresponding probability using the normal probability table in the appendix.

$P(0 < Z < 0.76)$ is given in the row starting with 0.7 and in the column starting with .06 of the standard normal probability table in the appendix.

Since the standard normal distribution is symmetric about 0, we then obtain the probability:

$$\begin{aligned} P(-0.76 < Z < 0) &= P(0 < Z < 0.76) \\ &= 0.2764 \\ &= 27.64\% \end{aligned}$$

🎯 Result

$$0.2764 = 27.64\%$$

PROBLEM STATEMENT

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Step 1

Given:

$$P(Z < 1.67)$$

Determine the corresponding probability using the normal probability table in the appendix.

$P(0 < Z < 1.67)$ is given in the row starting with 1.6 and in the column starting with .07 of the standard normal probability table in the appendix.

$$\begin{aligned} P(Z < 1.67) &= P(Z < 0) + P(0 < Z < 1.67) \\ &= 0.5000 + 0.4525 \\ &= 0.9525 \\ &= 95.25\% \end{aligned}$$

🎯 Result

$$0.9525 = 95.25\%$$

Chapter 7 Exercise 12

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PROBLEM STATEMENT

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Step 1

Two events are **independent**, if the probability that one event occurs in no way affects the probability of the other event occurring.

Result

Independent

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PROBLEM STATEMENT

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Step 1

Two events are **disjoint** or **mutually exclusive**, if the events cannot occur at the same time.

Result

Mutually exclusive

PROBLEM STATEMENT

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Step 1

An asymptotic distribution is a probability distribution that is the limiting distribution of a sequence of distributions. Since normal distributions are the limiting distribution for many other distributions and thus normal distributions are asymptotic.

Normal distributions are considered to be a family of distributions, because you obtain a different normal distribution when you have a different mean and/or a different standard deviation.

A normal distribution is also symmetric about its mean and thus 50% of the observations in a normal distribution are greater than the mean.

The property that does not hold for a normal probability distribution is then "only two outcomes," which is true as a normal probability distribution has infinitely many possible outcomes.

Result

Only two outcomes

PROBLEM STATEMENT

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Step 1

A normal probability distribution has a peak centered about the mean, while the normal probability distributions approaches zero on either side of the mean.

Moreover, this particular shape is called a bell-shape and thus the shape of a normal probability distribution is **bell-shaped**.

Result

Bell-shaped